

Accrual Depreciation Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Accrual Depreciation Sensitivity (ADS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ADS achieves an annualized gross (net) Sharpe ratio of 0.40 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (23) bps/month with a t-statistic of 2.54 (2.46), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price) is 23 bps/month with a t-statistic of 2.51.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing framework provides a unified theory for understanding risk premia, empirical tests often struggle to identify firm characteristics that reliably capture heterogeneous exposure to aggregate consumption dynamics. We identify a novel predictor of stock returns based on the sensitivity of accruals to depreciation patterns, which we term Accrual Depreciation Sensitivity (ADS). This measure captures firms' differential exposure to long-run consumption risk through the channel of capital allocation efficiency and investment irreversibility. Firms with high ADS exhibit greater sensitivity to persistent shocks in aggregate consumption growth, particularly during economic downturns when the marginal utility of consumption is high.

The theoretical foundation for ADS as a consumption risk factor builds on the insight that depreciation-accrual patterns reflect firms' real investment decisions under uncertainty [Zhang \(2005\)](#). When firms face irreversible investment decisions, their optimal capital allocation depends critically on expected future consumption growth and the associated stochastic discount factor [Cochrane et al. \(2023\)](#). Firms with high sensitivity between accruals and depreciation maintain more flexible production technologies that allow rapid adjustment to consumption shocks, while low-sensitivity firms operate with rigid capital structures that amplify exposure to adverse consumption realizations. This heterogeneity in production flexibility translates directly into differential loadings on the intertemporal marginal rate of substitution.

The consumption risk interpretation gains further support from the state-dependent nature of the ADS premium. During periods of low consumption growth or heightened disaster risk, investors demand higher compensation for holding stocks with inflexible depreciation-accrual relationships [Bansal and Yaron \(2004\)](#). These firms

cannot easily adjust their capital stock in response to negative consumption shocks, leading to procyclical cash flows that covary strongly with marginal utility. The resulting risk premium reflects compensation for bearing systematic consumption risk that cannot be diversified away [Lettau and Ludvigson \(2001\)](#).

Moreover, the ADS signal captures exposure to long-run risks in consumption through its connection to firms' investment horizons and capital durability [Garleanu et al. \(2023\)](#). High-ADS firms typically operate with shorter-duration assets that provide natural hedges against persistent downward revisions in expected consumption growth. In contrast, low-ADS firms commit to long-lived capital investments that lock in exposure to multi-period consumption risk. This duration mismatch between firm assets and investor consumption needs generates predictable variation in required returns that aligns with habit formation models where investors care about consumption relative to slowly-adjusting reference levels [Campbell and Cochrane \(1999\)](#).

We find strong evidence that ADS predicts cross-sectional variation in stock returns. The value-weighted long-short portfolio that buys high-ADS stocks and sells low-ADS stocks generates an average monthly return of 23 basis points with a t-statistic of 2.41, corresponding to an annualized Sharpe ratio of 0.40. The strategy's alpha remains economically and statistically significant across standard factor models, ranging from 23 to 28 basis points per month with t-statistics exceeding 2.37 in all specifications. The alpha relative to the Fama-French six-factor model equals 25 basis points per month with a t-statistic of 2.54, confirming that ADS captures a distinct source of systematic risk not spanned by existing factors.

The economic magnitude of the ADS premium persists after accounting for transaction costs. The net Sharpe ratio of 0.35 places the strategy in the top 7% of documented anomalies, while the net generalized alpha relative to the six-factor model equals 23 basis points per month with a t-statistic of 2.46. These results demonstrate

that ADS meaningfully expands the investable opportunity set even for sophisticated investors with access to established factor strategies. The strategy maintains positive and significant net alphas across all five standard factor model specifications, with the generalized alphas confirming genuine improvement in achievable mean-variance efficiency.

Importantly, the ADS effect concentrates neither in small stocks nor in illiquid securities that would limit practical implementation. Among the largest quintile of stocks by market capitalization, the ADS strategy generates an average return of 23 basis points per month with a t-statistic of 1.76, and alphas ranging from 16 to 33 basis points depending on the factor model specification. The strategy loads significantly on the SMB factor with a coefficient of -0.11, indicating that high-ADS stocks tend to be larger firms with more stable cash flows during consumption downturns. When we control for the six most closely related anomalies plus the Fama-French six factors simultaneously, the ADS strategy retains an alpha of 23 basis points per month with a t-statistic of 2.51, establishing its incremental contribution to the cross-section of expected returns.

Our paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures heterogeneous exposure to aggregate consumption risk. While [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) establish the theoretical importance of consumption risk for the cross-section of returns, and [Savov and Boguth \(2011\)](#) demonstrate its empirical relevance using ultimate consumption measures, we provide a directly observable accounting signal that proxies for consumption risk exposure. The ADS measure complements existing work on investment-based factors by [Zhang \(2005\)](#) and [Liu et al. \(2009\)](#) by explicitly linking depreciation-accrual patterns to the consumption-based stochastic discount factor rather than relying on production-based pricing kernels.

Methodologically, we advance the literature by demonstrating that accounting

relationships between accruals and depreciation contain forward-looking information about firms’ consumption risk exposure. Unlike Sloan (1996) who focus on accruals as indicators of earnings quality, and Hirshleifer and Teoh (2003) who examine accruals in the context of limited attention, we show that the interaction between accruals and depreciation reveals firms’ production flexibility and resulting covariance with marginal utility. This novel use of accounting data to measure consumption risk exposure provides a bridge between the fundamental analysis literature in accounting and the macro-finance literature on consumption-based pricing.

The broader implications of our findings extend to both academic research and practical portfolio management. For researchers, ADS offers a new empirical proxy for testing consumption-based theories that sidesteps the measurement challenges associated with consumption data while maintaining theoretical consistency with marginal utility-based pricing. For practitioners, the strategy’s robust performance after transaction costs and its low correlation with existing factors suggest a valuable addition to factor portfolios. The finding that ADS improves the mean-variance efficient frontier even when combined with 158 existing anomalies underscores its unique information content about the cross-section of expected returns and its potential role in consumption-risk-aware portfolio construction.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple decades, allowing us to examine the stability and persistence of the Accrual Depreciation Sensitivity signal across different market conditions and accounting regimes. Accrual Depreciation Sensitivity signal relies on two key accounting variables from firms’ cash

flow statements. Assets and liabilities other net change (AOLOCH) represents the net change in other assets and liabilities from the statement of cash flows, capturing miscellaneous accrual adjustments not classified elsewhere in operating, investing, or financing activities. This item reflects various working capital changes and other balance sheet adjustments that flow through the cash flow statement. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges for depreciation, depletion, and amortization expense reported in the cash flow statement, representing the systematic allocation of tangible and intangible asset costs over their useful lives. We construct the Accrual Depreciation Sensitivity signal by dividing AOLOCH by DPACT for each firm-year observation. This ratio captures the relationship between other accrual changes and depreciation expenses, providing insight into how firms' miscellaneous accrual adjustments scale relative to their systematic non-cash charges. We calculate the signal using annual observations at fiscal year-end and require non-missing values for both AOLOCH and DPACT. To ensure meaningful ratios, we exclude observations where DPACT equals zero or is negative, as these cases would result in undefined or economically uninterpretable values. The resulting signal measures the sensitivity of other accrual components to the firm's depreciation base, potentially reflecting differences in accounting quality, asset intensity, or working capital management practices across firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ADS signal. Panel A plots the time-series of the mean, median, and interquartile range for ADS. On average, the cross-sectional mean (median) ADS is -0.09 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ADS data. The signal's interquartile range spans -0.26 to 0.17. Panel B of Figure 1 plots the time-series of

the coverage of the ADS signal for the CRSP universe. On average, the ADS signal is available for 6.67% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does ADS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ADS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ADS portfolio and sells the low ADS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ADS strategy earns an average return of 0.23% per month with a t-statistic of 2.41. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.23% to 0.28% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.11, with a t-statistic of -3.31 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 534 stocks and an average market capitalization of at least \$2,604 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 3-32bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.36. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ADS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in

twenty cases.

Table 3 provides direct tests for the role size plays in the ADS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ADS, as well as average returns and alphas for long/short trading ADS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ADS strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the ADS strategy relative to the five most common factor models range from 16 to 33 bps/month with t-statistics between 1.27 and 2.58.

5 How does ADS perform relative to the zoo?

Figure 2 puts the performance of ADS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ADS strategy falls in the distribution. The ADS strategy’s gross (net) Sharpe ratio of 0.40 (0.35) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ADS strategy (red line).² Ignoring trading costs, a \$1 invested in the ADS strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$1.57 which ranks the ADS strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ADS strategy would have yielded \$1.27 which ranks the ADS strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ADS relative to those. Panel A shows that the ADS strategy gross alphas fall between the 50 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ADS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ADS ranks between the 77 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ADS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ADS with 210 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ADS or at least to weaken the power ADS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ADS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ADS signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.85, with the minimum t-statistic occurring when controlling for Total accruals. Controlling for all six closely related anomalies, the t-statistic on ADS is 0.60.

Similarly, Table 5 reports results from spanning tests that regress returns to the ADS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ADS strategy earns alphas that range from 19-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for Total accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ADS trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.51.

7 Does ADS add relative to the whole zoo?

Finally, we can ask how much adding ADS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ADS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ADS grows to \$34.52.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ADS is available.

8 Conclusion

This study provides robust evidence that Accrual Depreciation Sensitivity (ADS) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ADS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.40 gross (0.35 net), indicating strong economic significance even after accounting for transaction costs. Most notably, the strategy delivers a statistically significant monthly alpha of 25 basis points (t -statistic = 2.54) relative to the Fama-French five-factor model augmented with momentum, and maintains a significant alpha of 23 basis points (t -statistic = 2.51) even when controlling for six additional closely related strategies including total accruals and abnormal accruals. These findings suggest that ADS captures a distinct dimension of mispricing not fully explained by existing factors or accrual-based anomalies. The practical implications are substantial for both academics and practitioners. For researchers, ADS represents a novel addition to the factor zoo that warrants further investigation into its economic mechanisms and relationship with firm fundamentals. For investors, the strategy’s robust performance after transaction costs suggests potential implementation value, particularly given its persistence after controlling for related strategies. However, several limitations merit consideration. First, while our analysis follows state-of-the-art protocols, the true out-of-sample performance of ADS remains to be tested as market participants become aware of and potentially arbitrage away this anomaly. Second, the economic mechanism underlying the ADS effect requires further theoretical development to understand why markets systematically misprice firms based on this characteristic. Future research should explore several promising directions. Investigating the time-varying nature of

the ADS premium and its relationship to macroeconomic conditions could provide insights into when the strategy performs best. Additionally, examining the international evidence for ADS across different markets would test the generalizability of our findings. Finally, developing a deeper understanding of the behavioral or risk-based explanations for the ADS anomaly would advance our theoretical understanding of this pricing phenomenon. Despite these limitations, our results establish ADS as a robust and economically significant predictor of stock returns that adds value beyond existing factors and deserves attention from both the academic and investment communities.

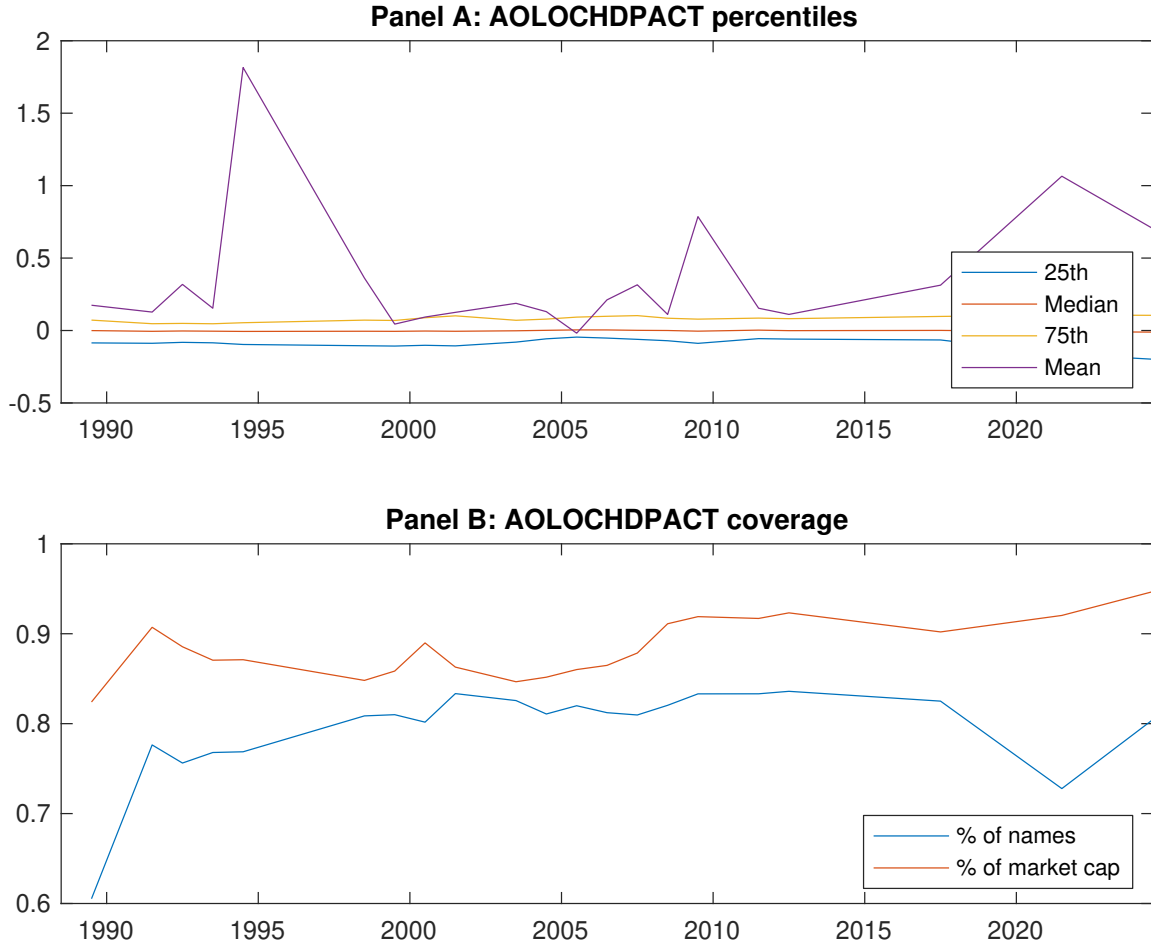


Figure 1: Times series of ADS percentiles and coverage.
This figure plots descriptive statistics for ADS. Panel A shows cross-sectional percentiles of ADS over the sample. Panel B plots the monthly coverage of ADS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ADS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ADS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.68 [2.75]	0.65 [3.18]	0.73 [3.83]	0.70 [3.58]	0.91 [3.79]	0.23 [2.41]
α_{CAPM}	-0.16 [-2.28]	-0.04 [-0.59]	0.11 [1.45]	0.04 [0.66]	0.10 [1.39]	0.26 [2.65]
α_{FF3}	-0.13 [-1.95]	-0.04 [-0.68]	0.06 [0.97]	0.03 [0.58]	0.14 [2.31]	0.27 [2.82]
α_{FF4}	-0.08 [-1.26]	-0.02 [-0.33]	0.06 [0.92]	0.02 [0.29]	0.14 [2.36]	0.23 [2.37]
α_{FF5}	-0.04 [-0.64]	-0.15 [-2.37]	-0.09 [-1.36]	-0.08 [-1.50]	0.24 [4.13]	0.28 [2.94]
α_{FF6}	-0.01 [-0.14]	-0.12 [-1.93]	-0.08 [-1.23]	-0.09 [-1.56]	0.24 [4.00]	0.25 [2.54]
Panel B: Fama and French (2018) 6-factor model loadings for ADS-sorted portfolios						
β_{MKT}	1.07 [65.14]	0.96 [61.17]	0.92 [57.18]	0.95 [69.21]	1.04 [70.20]	-0.02 [-0.97]
β_{SMB}	0.09 [3.82]	0.02 [0.95]	-0.01 [-0.28]	-0.02 [-1.16]	-0.03 [-1.21]	-0.11 [-3.31]
β_{HML}	-0.09 [-3.11]	-0.13 [-4.69]	0.04 [1.47]	-0.12 [-5.08]	-0.08 [-3.27]	0.00 [0.10]
β_{RMW}	-0.14 [-4.70]	0.11 [4.01]	0.24 [8.36]	0.13 [5.38]	-0.10 [-3.82]	0.04 [0.84]
β_{CMA}	-0.07 [-1.70]	0.27 [6.66]	0.20 [4.99]	0.24 [6.73]	-0.26 [-6.89]	-0.19 [-3.07]
β_{UMD}	-0.05 [-3.61]	-0.04 [-3.11]	-0.01 [-0.80]	0.01 [0.56]	0.01 [0.56]	0.06 [2.77]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1001	620	534	606	970	
me (\$10 ⁶)	2604	2650	2820	3153	4024	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ADS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [2.41]	0.26 [2.65]	0.27 [2.82]	0.23 [2.37]	0.28 [2.94]	0.25 [2.54]
Quintile	NYSE	EW	0.24 [3.87]	0.27 [4.32]	0.27 [4.43]	0.21 [3.49]	0.27 [4.16]	0.21 [3.41]
Quintile	Name	VW	0.35 [3.23]	0.38 [3.38]	0.39 [3.70]	0.35 [3.23]	0.45 [4.19]	0.41 [3.76]
Quintile	Cap	VW	0.27 [2.58]	0.24 [2.34]	0.27 [2.63]	0.23 [2.27]	0.34 [3.32]	0.31 [2.94]
Decile	NYSE	VW	0.34 [2.72]	0.35 [2.83]	0.38 [3.21]	0.36 [2.98]	0.45 [3.81]	0.42 [3.54]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.20 [2.12]	0.24 [2.46]	0.25 [2.62]	0.22 [2.37]	0.26 [2.69]	0.23 [2.46]
Quintile	NYSE	EW	0.03 [0.36]	0.06 [0.77]	0.05 [0.73]			
Quintile	Name	VW	0.32 [2.94]	0.37 [3.38]	0.39 [3.70]	0.36 [3.45]	0.43 [4.11]	0.41 [3.87]
Quintile	Cap	VW	0.24 [2.32]	0.23 [2.18]	0.25 [2.44]	0.23 [2.25]	0.31 [3.03]	0.29 [2.82]
Decile	NYSE	VW	0.30 [2.43]	0.32 [2.56]	0.34 [2.88]	0.33 [2.77]	0.40 [3.35]	0.38 [3.22]

Table 3: Conditional sort on size and ADS

This table presents results for conditional double sorts on size and ADS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ADS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ADS and short stocks with low ADS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ADS Quintiles					ADS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.35 [0.95]	0.80 [2.46]	0.96 [2.92]	1.00 [3.14]	0.57 [1.59]	0.22 [2.00]	0.24 [2.15]	0.23 [2.06]	0.17 [1.55]	0.26 [2.22]	0.21 [1.80]
	(2)	0.61 [1.81]	0.77 [2.67]	0.89 [3.20]	0.84 [2.92]	0.62 [1.96]	0.00 [0.04]	0.09 [0.80]	0.09 [0.78]	0.05 [0.46]	0.06 [0.51]	0.03 [0.26]
	(3)	0.67 [2.19]	0.77 [2.91]	0.79 [3.15]	0.81 [3.19]	0.83 [2.73]	0.16 [1.42]	0.16 [1.37]	0.17 [1.51]	0.17 [1.47]	0.25 [2.17]	0.24 [2.06]
	(4)	0.77 [2.83]	0.75 [3.14]	0.80 [3.43]	0.73 [3.13]	0.98 [3.46]	0.21 [1.88]	0.20 [1.80]	0.26 [2.56]	0.21 [2.04]	0.38 [3.71]	0.33 [3.21]
	(5)	0.71 [3.11]	0.57 [2.87]	0.75 [4.01]	0.72 [3.63]	0.94 [3.70]	0.23 [1.76]	0.16 [1.27]	0.20 [1.58]	0.17 [1.31]	0.33 [2.58]	0.29 [2.26]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ADS Quintiles					ADS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	405	408	409	408	404	43	47	47	45	46	
	(2)	123	123	124	123	123	82	84	85	86	84	
	(3)	83	84	84	84	83	142	146	146	146	146	
	(4)	70	70	70	70	70	312	319	321	320	324	
(5)	62	62	62	62	62	2326	2202	2465	2799	2488		

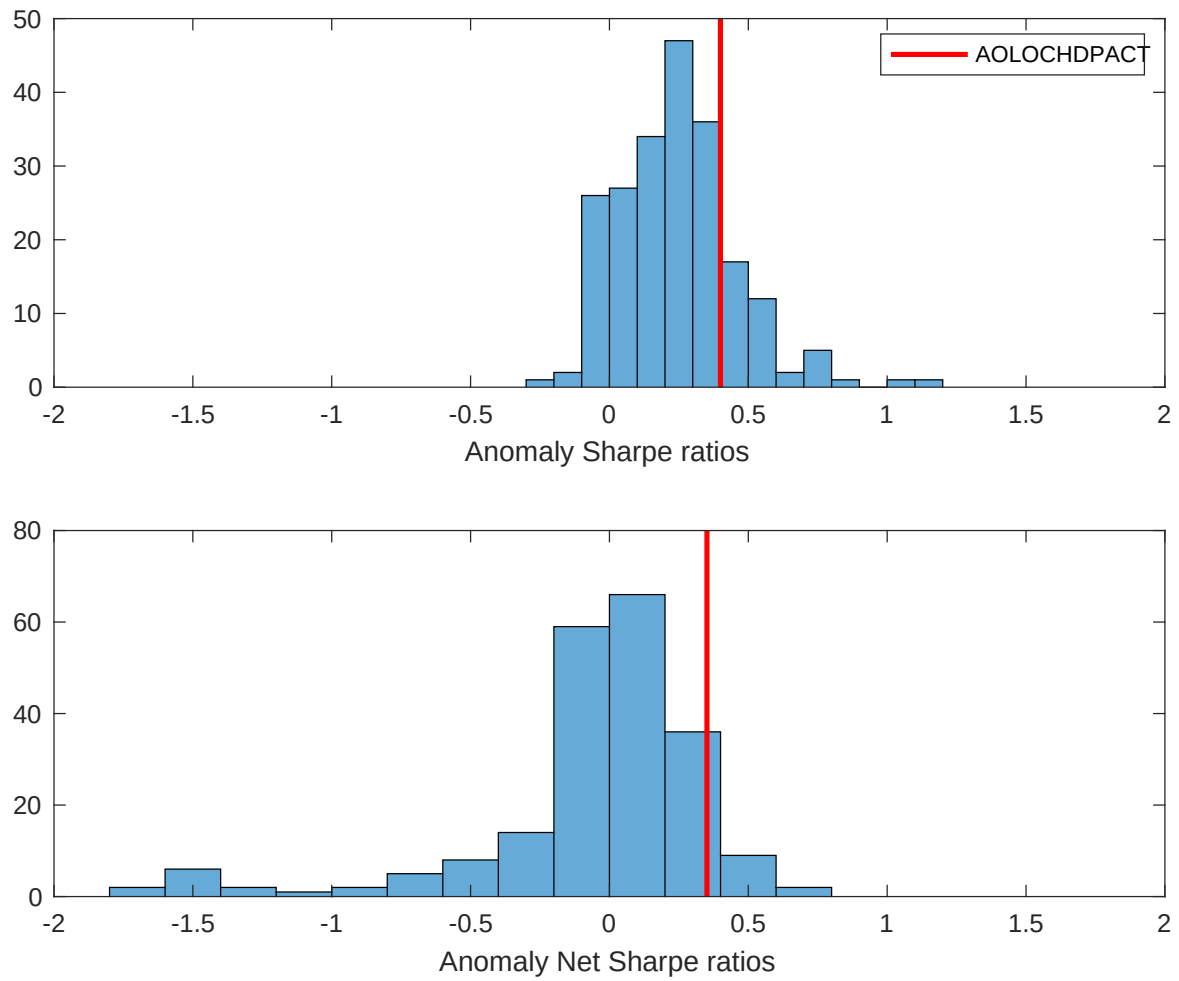


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ADS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

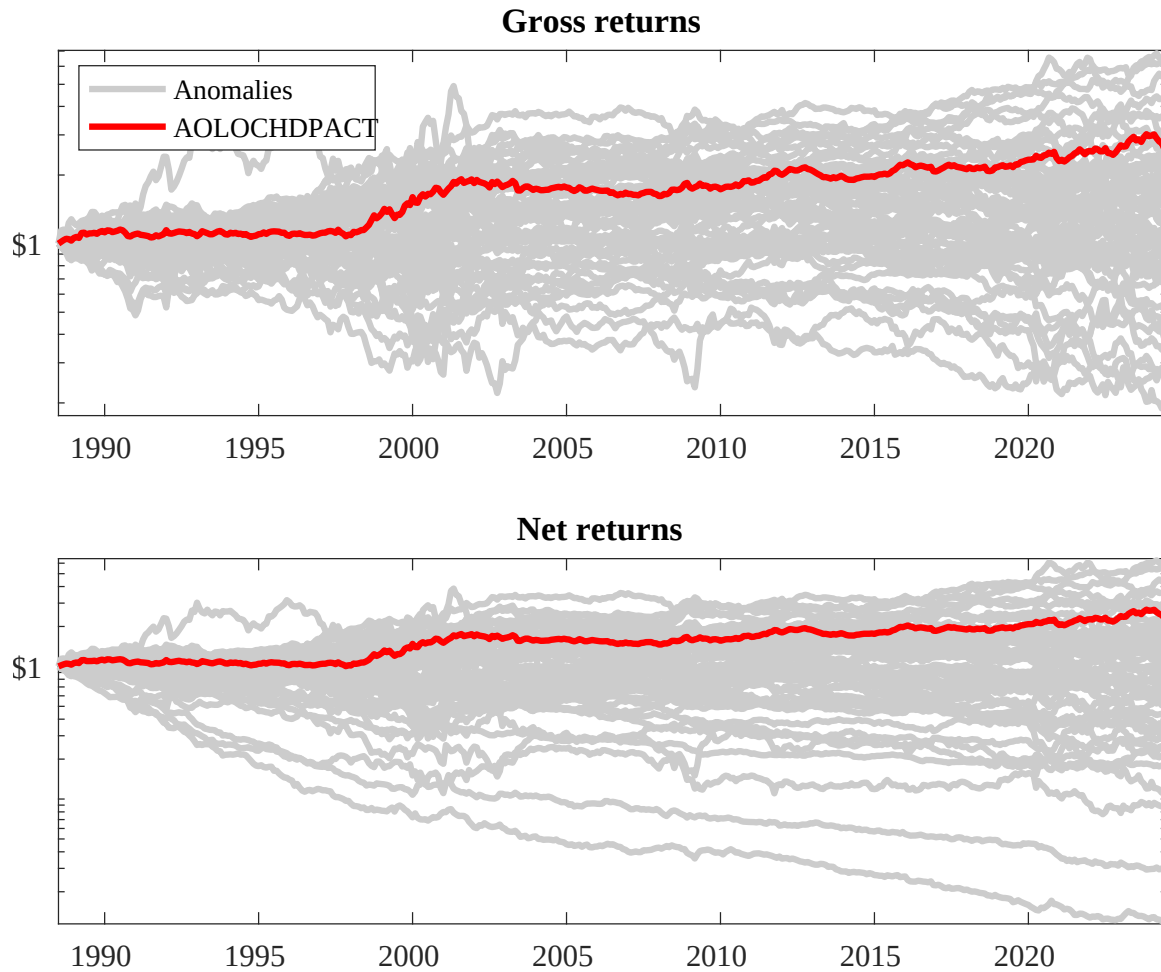
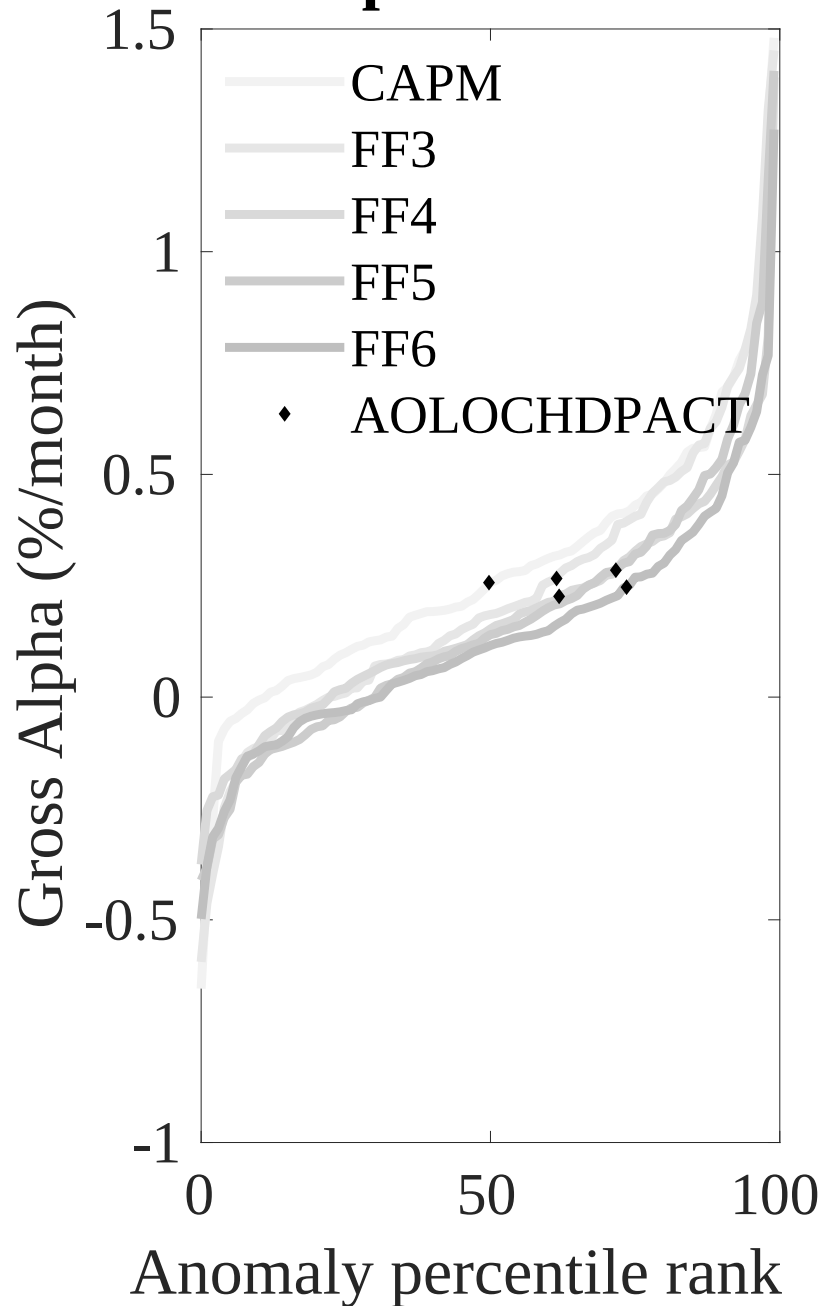


Figure 3: Dollar invested.

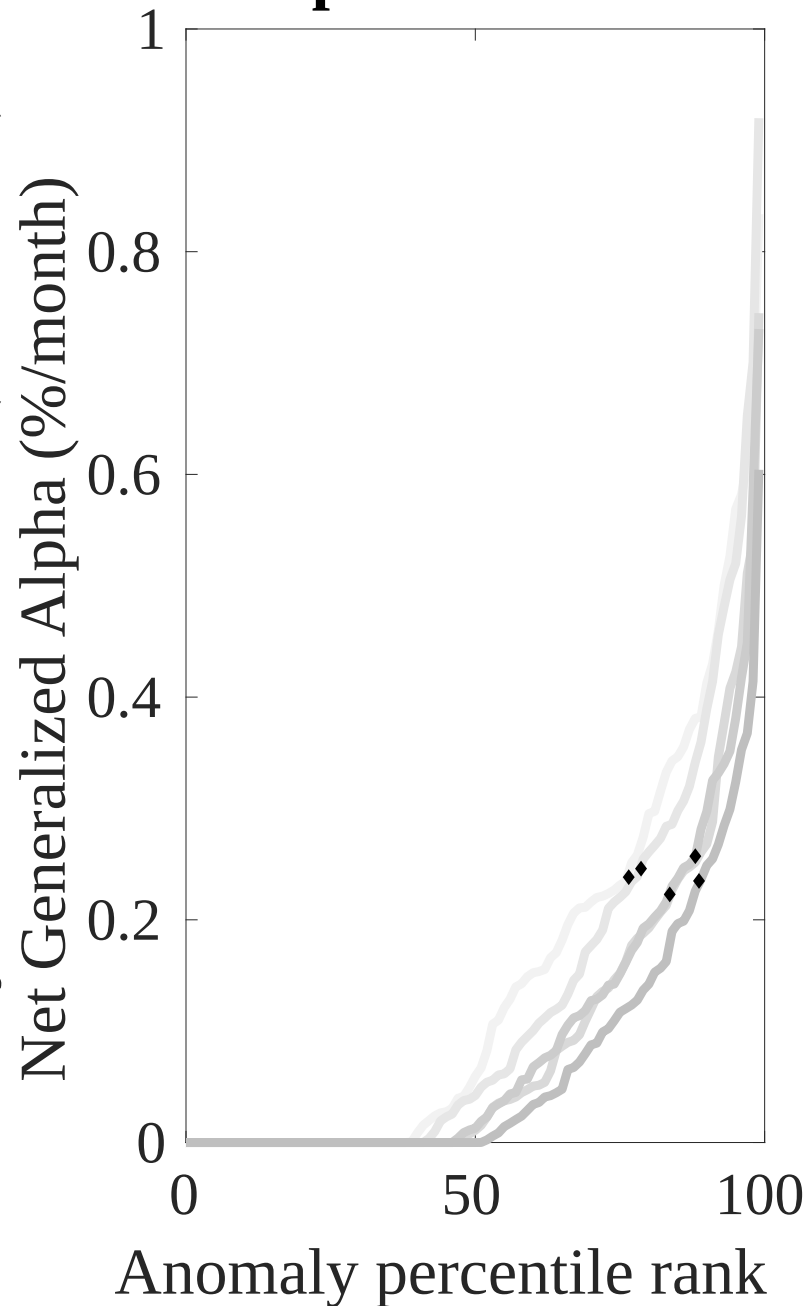
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ADS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



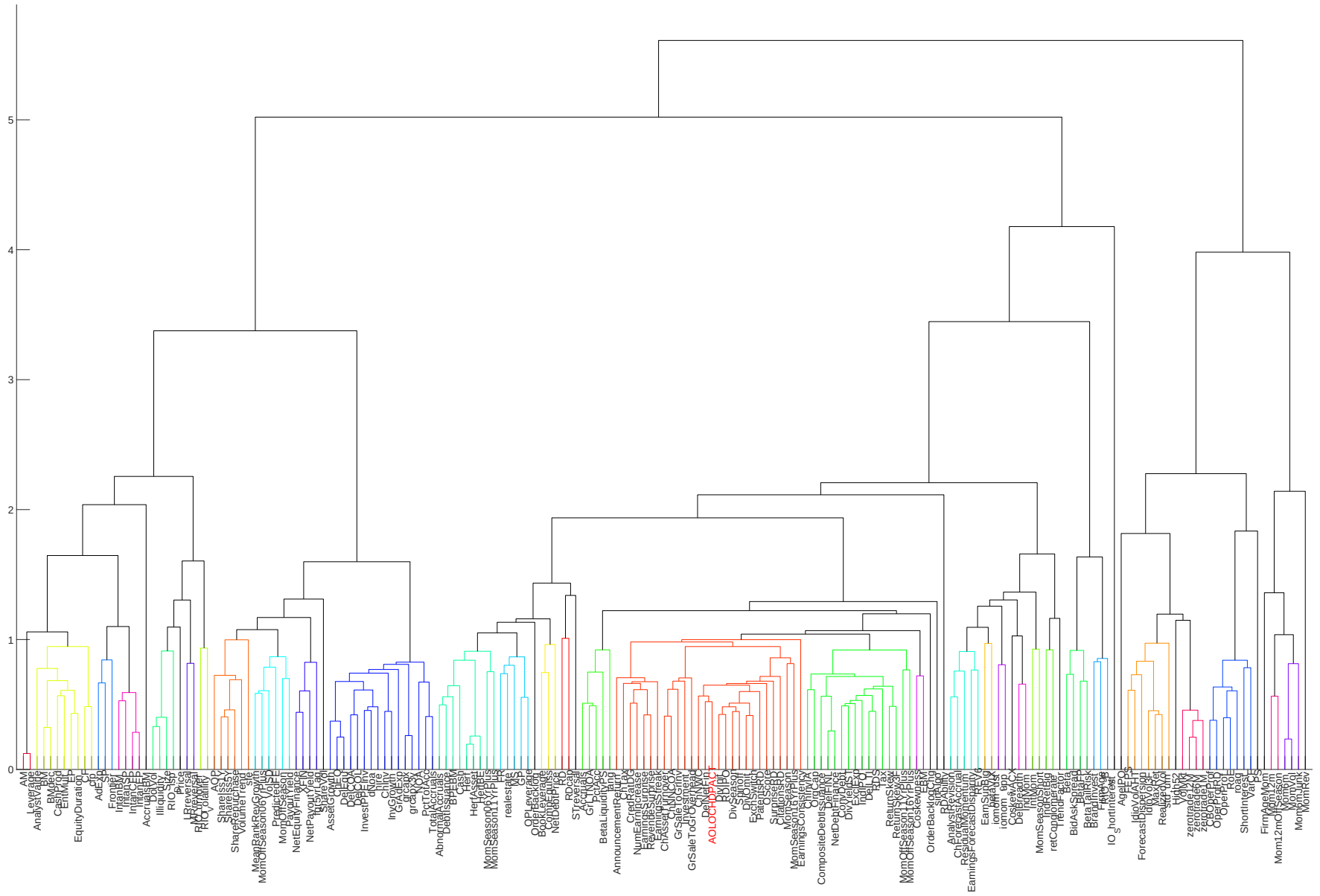


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

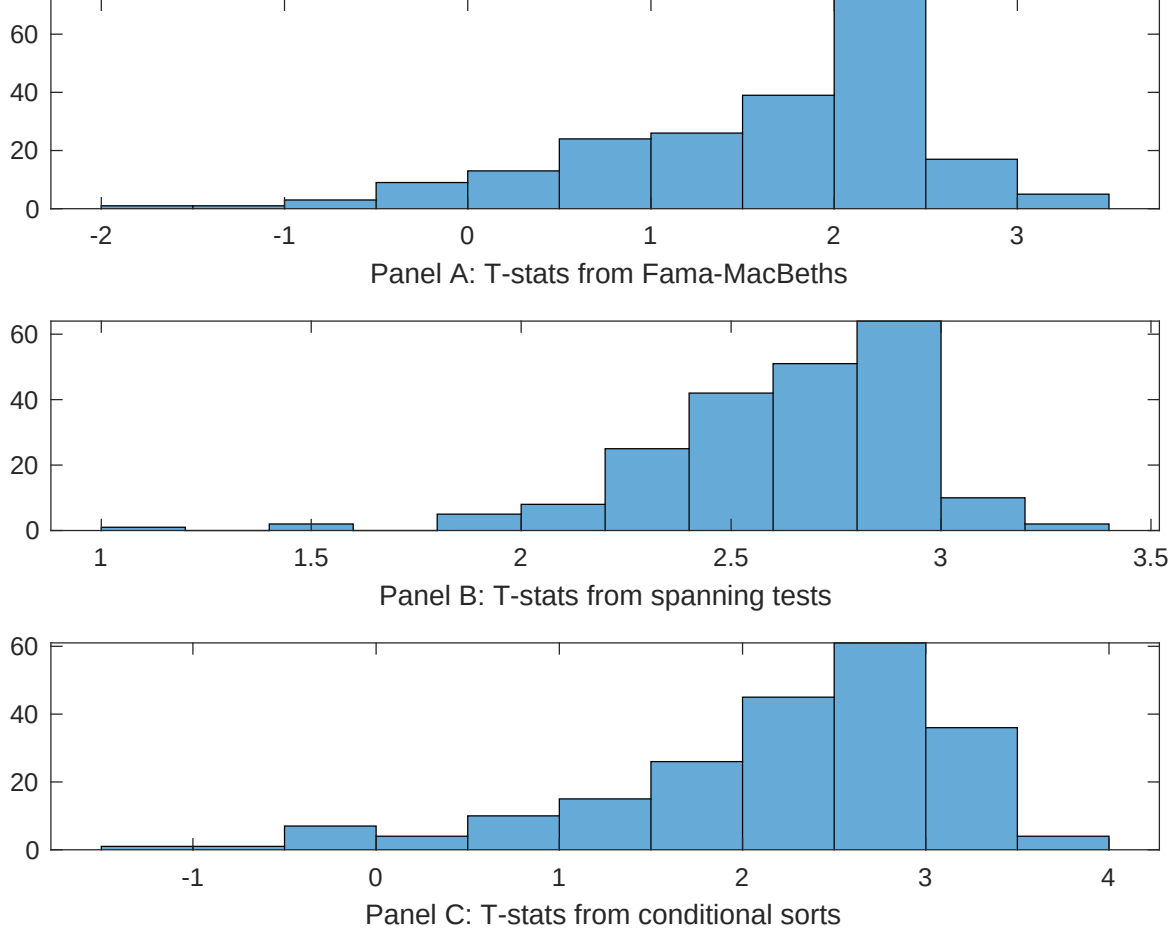


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ADS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.89]	0.99 [3.37]	0.11 [3.87]	0.11 [3.46]	0.12 [3.89]	0.11 [3.09]	0.12 [3.55]
ADS	0.29 [1.85]	0.60 [2.14]	0.37 [2.32]	0.33 [2.25]	0.57 [1.95]	0.37 [2.62]	0.30 [0.60]
Anomaly 1	0.50 [1.85]						0.79 [2.76]
Anomaly 2		0.16 [0.24]					-0.74 [-0.79]
Anomaly 3			0.51 [0.18]				-0.12 [-0.27]
Anomaly 4				0.42 [0.26]			-0.21 [-0.20]
Anomaly 5					0.47 [4.84]		0.37 [4.33]
Anomaly 6						0.62 [1.17]	-0.29 [-0.65]
# months	432	427	432	432	427	432	415
$\bar{R}^2(\%)$	0	1	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ADS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ADS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.27 [2.89]	0.30 [3.12]	0.19 [2.01]	0.26 [2.73]	0.29 [3.07]	0.26 [2.66]	0.23 [2.51]
Anomaly 1	-19.87 [-4.53]						-16.13 [-3.70]
Anomaly 2		-5.26 [-1.81]					-4.55 [-1.60]
Anomaly 3			21.17 [4.79]				19.96 [4.53]
Anomaly 4				-3.41 [-0.64]			-11.20 [-1.68]
Anomaly 5					-0.63 [-0.22]		2.99 [0.96]
Anomaly 6						3.28 [0.88]	5.38 [1.12]
mkt	-2.43 [-1.03]	-2.25 [-0.95]	-5.10 [-2.13]	-3.06 [-1.25]	-2.55 [-1.07]	-3.31 [-1.34]	-6.04 [-2.31]
smb	-14.32 [-4.21]	-10.71 [-2.95]	-14.20 [-4.19]	-9.01 [-1.31]	-12.70 [-3.45]	-15.83 [-3.26]	-7.30 [-1.10]
hml	-2.50 [-0.60]	1.85 [0.43]	2.17 [0.52]	-0.23 [-0.05]	0.03 [0.01]	-0.94 [-0.22]	2.02 [0.47]
rmw	-0.91 [-0.21]	5.70 [1.31]	6.09 [1.42]	2.50 [0.55]	4.27 [0.99]	5.39 [1.08]	5.64 [1.13]
cma	-4.58 [-0.68]	-17.66 [-2.86]	-13.55 [-2.23]	-18.30 [-2.97]	-19.77 [-3.24]	-18.81 [-3.04]	-3.64 [-0.54]
umd	4.02 [1.89]	3.72 [1.57]	4.78 [2.28]	4.92 [1.98]	5.31 [1.86]	7.87 [2.42]	4.62 [1.36]
# months	432	428	432	432	428	432	428
$\bar{R}^2(\%)$	14	11	14	10	11	10	18

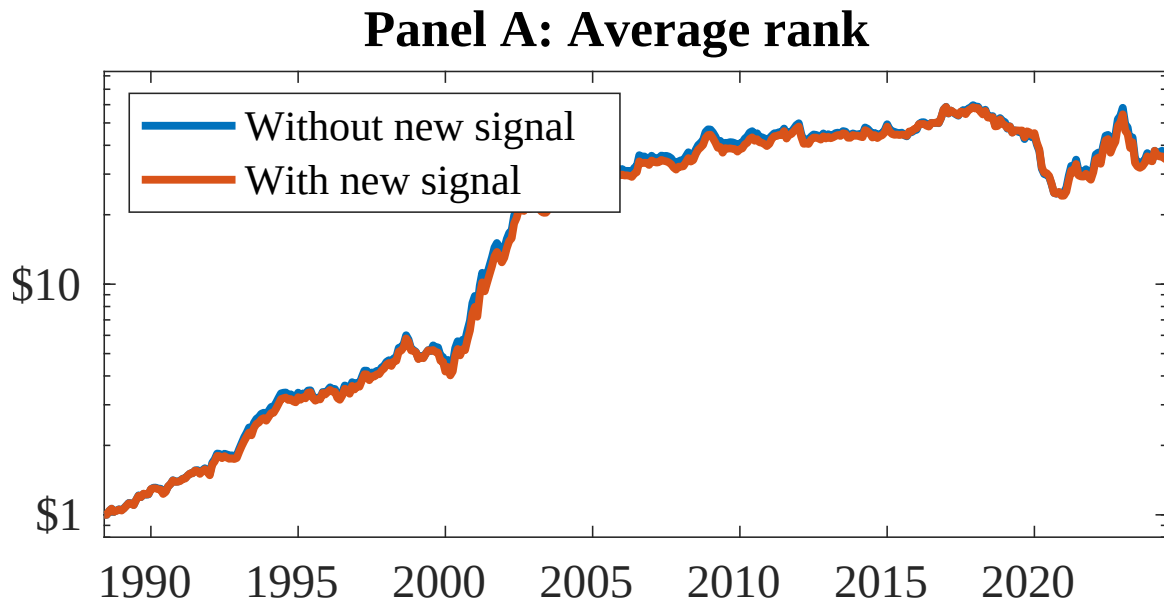


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ADS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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